MINISTRY OF SCIENCE AND HIGHER EDUCATION OF RUSSIAN FEDERATION
FEDERAL STATE AUTONOMOUS EDUCATIONAL INSTITUTION OF HIGHER EDUCATION
«PEOPLES' FRIENDSHIP UNIVERSITY OF RUSSIA NAMED AFTER PATRICE LUMUMBA »

Institute of Environmental Engineering

ANNOTATION

Graduation Qualification Work

<u>Name</u>

(**Stanley Chukwuebuka Okonkwo)**

on the topic: "**Title" Ecological Impacts of Urbanization on Green Spaces and Drainage Patterns in Lagos State, Nigeria: A GIS and Remote Sensing Approach"**

The research aim is to assess the ecological impacts of urbanization on green spaces and drainage patterns across all 20 Local Government Areas of Lagos State, Nigeria, for the period 2000–2021. The study applies a comprehensive GIS and remote sensing methodology integrating six globally validated open-access datasets: GHSL P2023A, ESA WorldCover 2021, Landsat 7/8, Sentinel-2 MSI, SRTM DEM, and JRC Global Surface Water, processed in Google Earth Engine and Python 3.10. Key findings include: urban built-up expansion of 41.1% from 2000 to 2020; 15 of 20 LGAs falling below the WHO minimum of 9 m^2 per capita green space; metropolitan drainage density of 0.20 km/km^2 against the internationally recommended 2.0 standard; and 3.77 million residents (33.8%) residing in High or Very High flood vulnerability zones. Statistical analysis confirmed imperviousness as the dominant driver of vegetation loss (Pearson $r = -0.78$, $p < 0.001$), with significant spatial clustering across all seven environmental variables (Global Moran’s $I = 0.304–0.490$, $p < 0.001$). Six evidence-based policy proposals are formulated for sustainable urban environmental management in Lagos State.

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